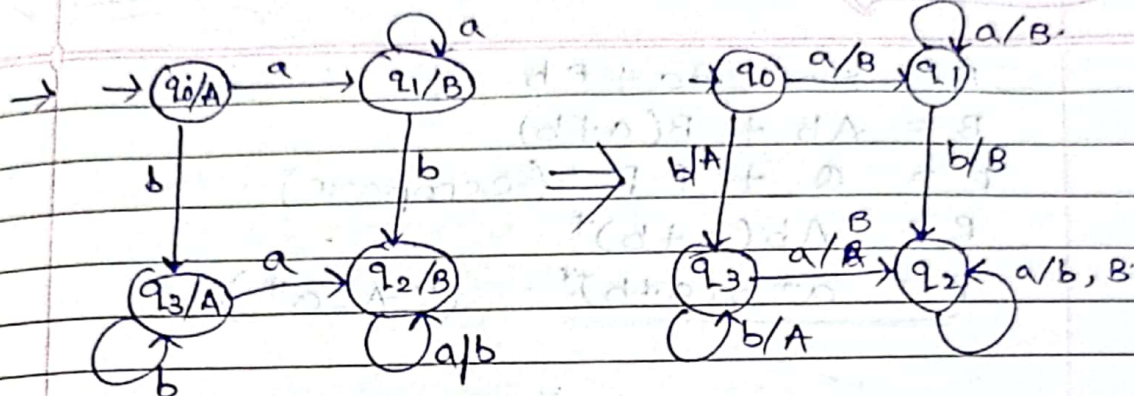


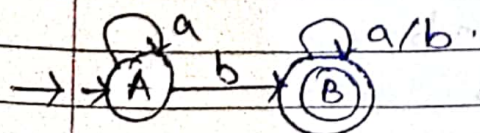


* ARDER'S LEMMA (METHOD) :-

~~This~~ This mechanism is used for only DFA & NFA. It cannot be used for e-NFA.

a) If P is free from ϵ then,
 $R = Q + RP$ then, $R = QP^*$

b) If P containing ϵ then,
 $R = Q + RP$ has infinite solution.



$$\underline{RE = a^*b(a+b)^*}$$

$$A \Rightarrow \epsilon + Aa$$

$$R \Rightarrow Q + RP \text{ (Compare)}$$

$$A = \epsilon a^* = \underline{a^*}$$

$$\begin{aligned} B &= Ab + Ba + Bb \\ B &= Ab + B(a+b) \\ R &= Q + RP \text{ (compare)} \\ R &= Ab(a+b)^* \\ \underline{R} &= \underline{a^*b(a+b)^*} \quad (\because A = a^*) \end{aligned}$$

→ Construct the RE. that generate all the string of 'a' & 'b' where every string do not end with.

$$L = \{ a, b, ab, ba, aab, bba, aba, \dots \}$$

~~X~~ RE = $(a+b)^*b + a + \epsilon$.



$$\begin{aligned} q_0 &= \epsilon + q_0b + q_1b + q_2b \\ \Rightarrow q_0 &= \epsilon + (q_0 + q_1 + q_2)b \end{aligned}$$

$$\begin{aligned} q_1 &= q_0a \\ q_2 &= q_1a + q_2a \\ R &= Q + RP \\ q_2 &= q_0a + q_1a \\ \therefore q_2 &= q_0aa^* \end{aligned}$$

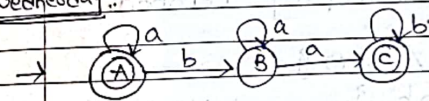
$$\begin{aligned} \therefore q_0 &= \epsilon + q_0b + q_0ab + q_0aaa^*b \\ \therefore q_0 &= \epsilon + q_0(b+ab+aaa^*b) \\ R &= Q + RP \end{aligned}$$

$$\begin{aligned} \therefore q_0 &= \epsilon + (q_0 + q_0a + q_0aa^*)b \\ \Rightarrow q_0 &= (b+ab+aaa^*b)^* \\ q_1 &= (b+ab+aaa^*b)^*a \end{aligned}$$

$$RE = q_0 + q_1$$

$$\therefore RE = (b+ab+aaa^*b)^*(1+a)$$

30/7/25
wednesday



$$RE = a^* + a^*ba^*ab^*$$

Using Arden's, $R = Q + RP$
 $A = \epsilon + Aa \Rightarrow A = \epsilon a^* \Rightarrow A = a^*$

$$\begin{aligned} B &= Ab + Ba = a^*b + Ba \\ \therefore B &= QP^* \therefore B = a^*ba^* \end{aligned}$$

$$\begin{aligned} C &= Ba + Cb = a^*ba^*a + Cb \\ \therefore C &= a^*ba^*ab^* \end{aligned}$$

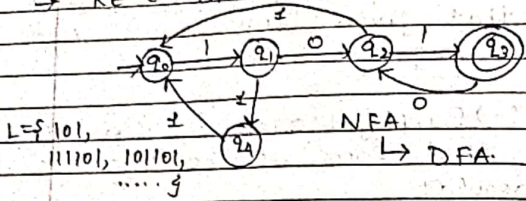
$$\begin{aligned} RE &= A + C \\ RE &= \underline{a^* + a^*ba^*ab^*} \end{aligned}$$

8/07/25.

Thursday.

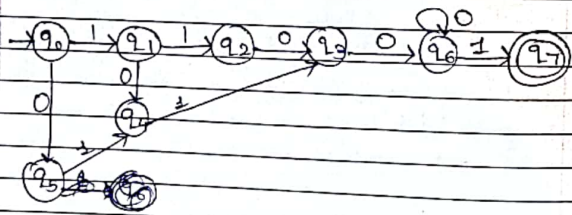
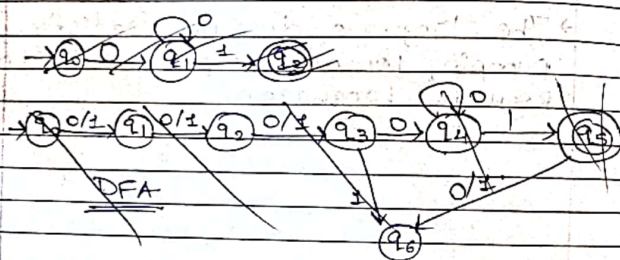
Date Page 62.

→ RE to DFA $(111+101)^*101$.

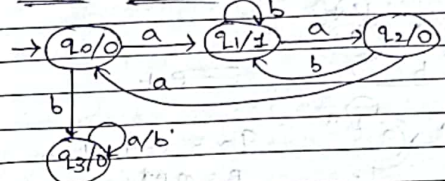


→ RE to NFA $(110+101+011) \cdot 00^*1$.

$L = \{11001, 10100^*1, 01100^*1, \dots\}$



→ Moore to Mealy

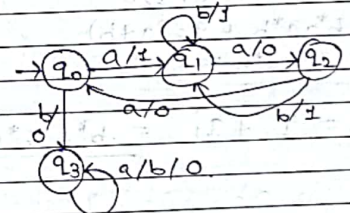


Moore Table.

	a	b	Δ
→ q ₀	q ₁	q ₃	0
q ₁	q ₂	q ₁	1
q ₂	q ₀	q ₁	0
q ₃	q ₃	q ₃	0

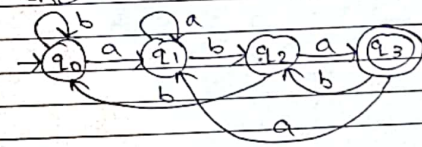
Mealy Table

	a	Δ	b	Δ
→ q ₀	q ₁	1	q ₃	0
q ₁	q ₂	0	q ₁	1
q ₂	q ₀	0	q ₁	1
q ₃	q ₃	0	q ₃	0

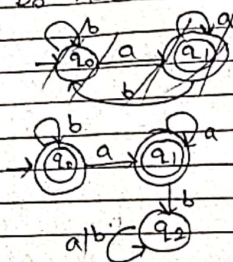


→ End 'aba'

$RE = (a+b)^* aba$



→ Do not subtracting 'ab'



$RE = a^+b^+$

$$q_0 = \epsilon + q_0b$$

$$R = aP^*$$

$$q_0 = b^*$$

$$q_1 = q_0a + q_1a$$

$$q_1 = b^*a + q_1a \Rightarrow q_1 = b^*aa^*$$

$$q_2 = q_1b + q_2a + q_2b$$

$$q_2 = b^*aa^*b + q_2(a+b)$$

$$\therefore q_2 = b^*aa^*b(a+b)^*$$

$$RE = q_0 + q_1 = b^* + b^*aa^*$$

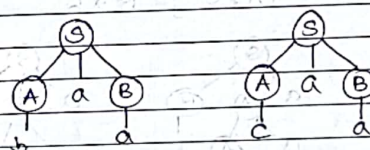
01/08/25
Friday

* GRAMMAR :- The set of all the production which are used to generate the strings is known as Grammar.

→ Grammar is of 4 tuple. (V, T, P, S) where, V = Variable, T = Terminal, P = Production, S = Start Symbol.

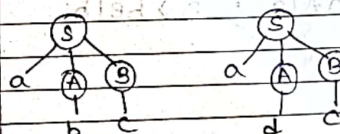
→ Grammar can be denoted by 'G'.

→ $S \rightarrow AaB$; $A \rightarrow b$; $B \rightarrow a$; $A \rightarrow c$.



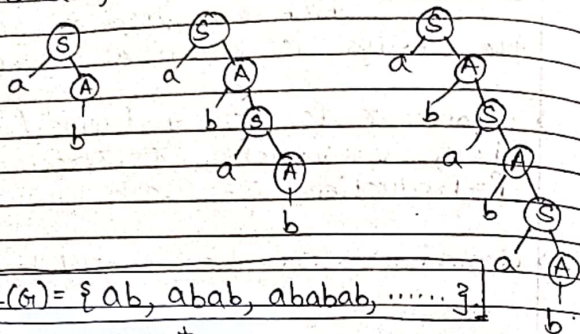
$$\therefore L(G) = \{baa, caa\}$$

→ $S \rightarrow aAB$; $A \rightarrow bld$; $B \rightarrow c$.



$$\therefore L(G) = \{abc, adc\}$$

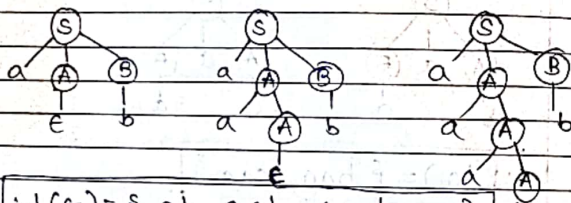
→ $S \rightarrow aA; A \rightarrow bS/b$



$\therefore L(G) = \{ab, abab, ababab, \dots\}$

$RE = (ab)^+$

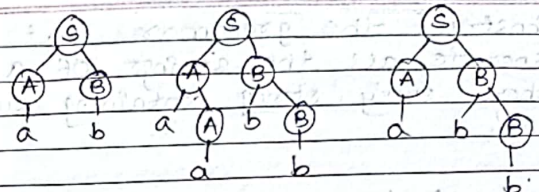
→ $S \rightarrow aAB; A \rightarrow aA/e; B \rightarrow b$



$\therefore L(G) = \{ab, aab, aaab, \dots\}$

$RE = a^+b$

→ $S \rightarrow AB; A \rightarrow aA/a; B \rightarrow bB/b$

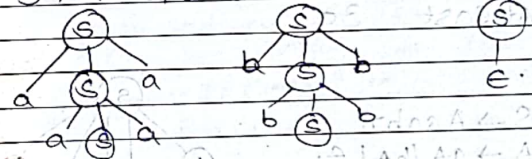


$L(G) = \{ab, aabb, abbb, \dots\}$

$RE = a^+b^+$

$L = \{a^n b^m / n, m \geq 1\}$

→ $S \rightarrow aSa / bSb / e$



$L(G) = \{e, aa, bb, abba, baab, aabbba, bbaabb, \dots\}$

$L = \{cw^R / \{a, b\}^*\}$

04/08/25

Language to Grammar

③ Construct the grammar that generate all the strings of a & b where every string containing substring 'aab'.

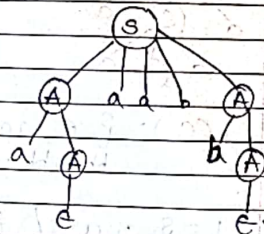
② Construct the grammar that generate all the strings of a & b where every string containing the 4th symbol from L.H.S. is always 'a'.

③ Almost $\exists a$.

④ Exact $\exists a$.

⑤ Atleast $\exists a$.

③ $S \rightarrow AaabaA$
 $A \rightarrow aA/bA/\epsilon$



④ $|W|_a = 3$ $S \rightarrow AaAaAaA$
 $A \rightarrow bA/\epsilon$

⑤ $|W|_a \geq 3$ $S \rightarrow AaAaAaA$
 $A \rightarrow aA/bA/\epsilon$

③ $|W|_a \leq 3$ $S \rightarrow BABABAB$
 $A \rightarrow a/\epsilon$
 $B \rightarrow bB/\epsilon$

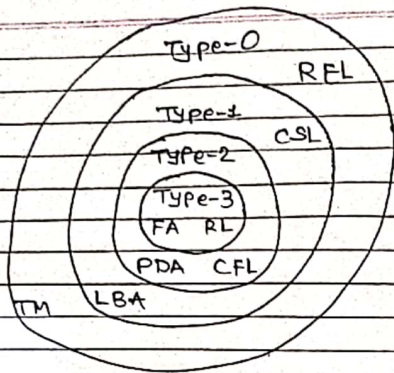
② $S \rightarrow AAAaB$
 $A \rightarrow a/b$
 $B \rightarrow aB/bB/\epsilon$

⑥ start & end with same Symbol.
 $\Sigma = \{0,1\}$
 $S \rightarrow ABA/CBC/A/C$
 $A \rightarrow 0$; $B \rightarrow 0B/1B/\epsilon$; $C \rightarrow 1$.

⑦ Start & end with different Symbol.

$S \rightarrow ABC/CBA$
 $A \rightarrow 0$; $B \rightarrow 0B/1B/\epsilon$; $C \rightarrow 1$.

07/08/25
Thursday.



* CHOMSKY HIERARCHY:-

→ Based on production grammar is divided into 4 types

⇒ Type-0 :- The grammar is said to be type-0 or Recursive Enumerable Grammar if every production is in the form of $\alpha \rightarrow \beta$ where, $\alpha, \beta \in (V+T)^*$.

NOTE:- Every grammar should contain atleast one production which has a non-terminal at left-hand-side.

⇒ Type-1 :- The grammar is said to be type-1 or Context Sensitive Grammar if every production is in the form of $\alpha \rightarrow \beta$ where, $|\alpha| \leq |\beta|$ & $\beta \neq \epsilon$ and $\alpha, \beta \in (V+T)^+$.

NOTE:- Context Sensitive Grammar is also called as "length increasing Grammar."

⇒ Type-2 :- The grammar is said to be type-2 or Context free Grammar if every production is in the form of $\alpha \rightarrow \beta$ where, $\beta \in (V+T)^*$ and $\alpha \in V$.

⇒ Type-3 :- The grammar is said to be type-3 or Regular Grammar if every production is in the form of $\alpha \rightarrow \beta X / X \beta$ where, $\alpha, \beta \in V$ and $X \in T$.

21/08/25
Thursday

(DPDA, NDPDA)

* PUSH DOWN AUTOMATA (PDA) :-

$(Q, \Sigma, \delta, q_0, f, \Gamma, Z_0)$

set of all stack symbol. $\rightarrow$ initial state of stack.

$Q \times \Sigma \times \Gamma \rightarrow Q \times \Gamma^*$

$Q \times \{ \epsilon, \Sigma \cup \Gamma \} \times \Gamma \rightarrow Q \times \Gamma^*$

1. Push

aabb

$(q_0, Z_0, a/aZ_0) \rightarrow (q_1, a)$

$\delta(q_0, a, Z_0) \Rightarrow (q_1, aZ_0)$

TF: $\rightarrow$

2. Pop

$\rightarrow bb$

$\rightarrow (q_1, b, a/e) \rightarrow (q_2, a)$

$\delta(q_1, b, a) \Rightarrow (q_2, \epsilon)$

3. skip

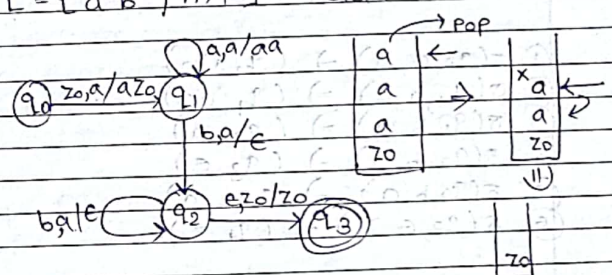
cbb

$(q_2, c, a/a) \rightarrow (q_3, a)$

$\delta(q_2, c, a) \Rightarrow (q_3, a)$

$L = \{ a^n b^n / n \geq 1 \}$

aaa bbb



1. $\delta(q_0, a, Z_0) \Rightarrow (q_1, aZ_0)$

2. $\delta(q_1, a, a) \Rightarrow (q_1, aa)$

3. $\delta(q_1, b, a) \Rightarrow (q_2, \epsilon)$

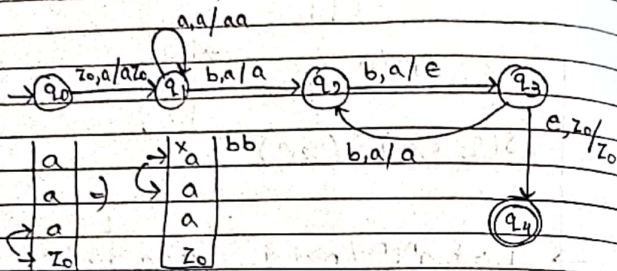
4. $\delta(q_2, b, a) \Rightarrow (q_2, \epsilon)$

5. $\delta(q_2, \epsilon, Z_0) \Rightarrow (q_3, Z_0)$

22/08/25
Friday

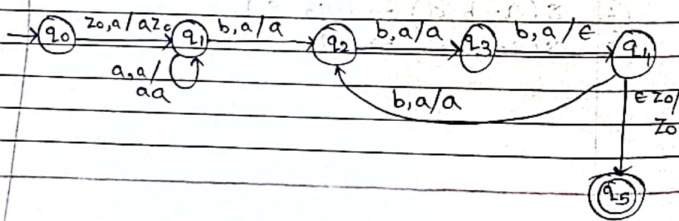
Date
Page 76

→ $L = \{ a^n b^{2n} \mid n \geq 1 \}$
At $n=3 \Rightarrow a a a b b b b b b$.



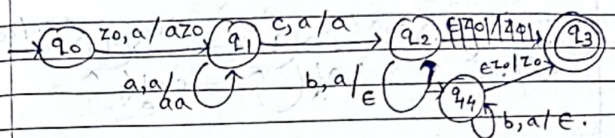
- 1) $\delta(q_0, z_0, a) \Rightarrow (q_1, a z_0)$
- 2) $\delta(q_1, a, a) \Rightarrow (q_1, aa)$
- 3) $\delta(q_1, b, a) \Rightarrow (q_2, a)$
- 4) $\delta(q_2, b, a) \Rightarrow (q_3, e)$
- 5) $\delta(q_3, b, a) \Rightarrow (q_2, a)$
- 6) $\delta(q_3, e, z_0) \Rightarrow (q_4, z_0)$

→ $L = \{ a^n b^{3n} \mid n \geq 1 \}$
At $n=3 \Rightarrow a a a b b b b b b b b b$.

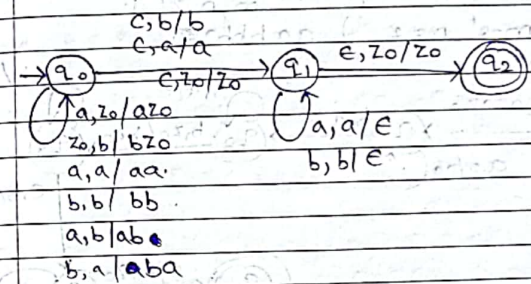


Date
Page 77

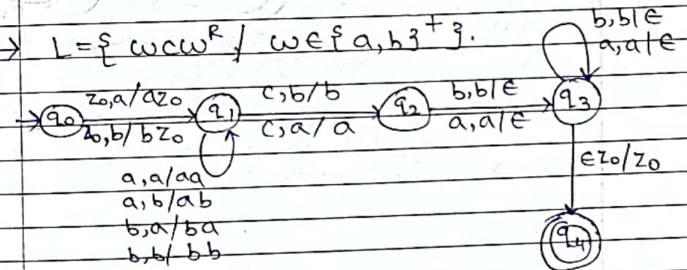
→ $L = \{ a^n c b^n \mid n \geq 1 \}$
At $n=2 \Rightarrow a a c b b$.



→ $L = \{ w c w^R \mid w \in \{a, b\}^* \}$

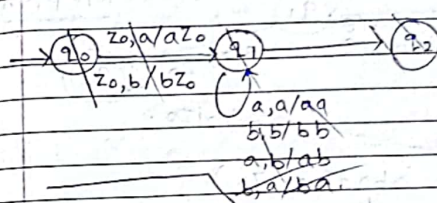
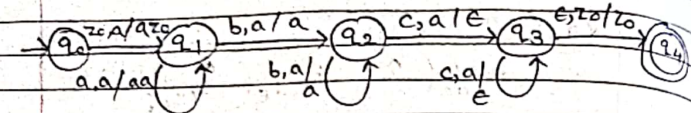


→ $L = \{ w c w^R \mid w \in \{a, b\}^+ \}$

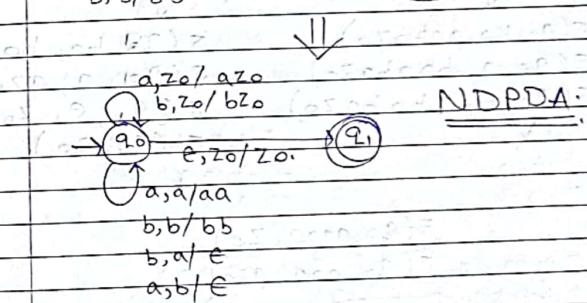
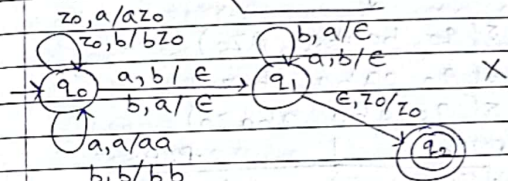


→ $L = \{a^n b^m c^n \mid m, n \geq 1\}$
 $m=1, n=2 \rightarrow aabcc$

→ $L = \{w \mid w_{a=b} \mid w \in \{a, b\}^+\}$
 $aabb$

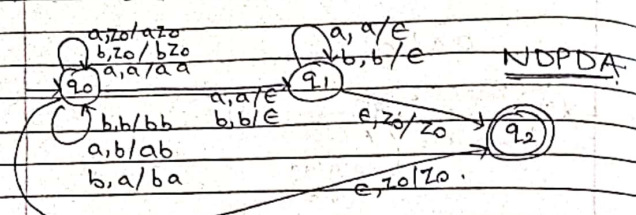


→ $L = \{a^n b^{m+n} c^m \mid m, n \geq 1\}$
 $m=1, n=2 \rightarrow aabbbcc$



NDPDA.

→ $L = \{ ww^R \mid w \in \{a, b\}^* \}$



$\delta(q_0, abaaba, z_0)$

$\delta(q_0, baaba, az_0)$

$\delta(q_0, aaba, baz_0)$

$\delta(q_0, aba, abaz_0)$

push | pop

$\delta(q_0, ba, aaba z_0)$

$\delta(q_0, ba, baz_0)$

$\delta(q_0, a, baaba z_0)$

$\delta(q_0, a, az_0)$

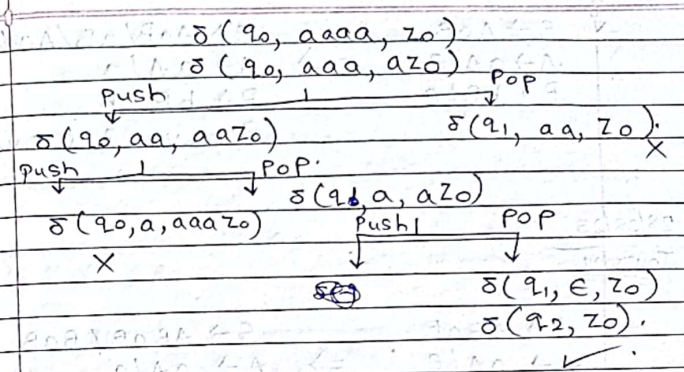
$\delta(q_0, \epsilon, ababaz_0)$

$\delta(q_1, \epsilon, z_0)$

X

$\delta(q_2, z_0)$

$\delta(q_0, aaaa, z_0)$
 $\delta(q_0, aaa, az_0)$



* CFG to CNF :-

step-1 :- Remove ϵ .

step-2 :- Remove Unit Production.

step-3 :- → Remove useless symbol.

→ Remove Dead Symbol.

After completing first 3-steps then,
step-4 :- Convert all Terminal into Variable. (exclude single Terminal).

$\rightarrow S \rightarrow AaB$
 $A \rightarrow aA/\epsilon$
 $B \rightarrow bB/\epsilon$

$\Rightarrow S \rightarrow AaB/aB/Aa/a.$
 $A \rightarrow aA/a$
 $B \rightarrow bB/b$
 Remove ϵ production.

28/08/25
Thursday.

$\rightarrow S \rightarrow ABaB$
 $A \rightarrow aA/\epsilon$
 $B \rightarrow bB/b$

$\Rightarrow S \rightarrow ABaB/BaB$
 $A \rightarrow aA/a$
 $B \rightarrow bB/b.$

$\rightarrow S \rightarrow ABaB$
 $A \rightarrow aA/\epsilon$
 $B \rightarrow bB/b/\epsilon$

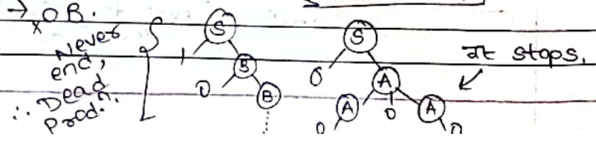
$\Rightarrow S \rightarrow ABaB/BaB/AaB/ABa/$
 $A \rightarrow aA/a$
 $B \rightarrow bB/b$
 Aa/Ba

$\rightarrow S \rightarrow A/B$
 $A \rightarrow aA/bB$
 $B \rightarrow b.$

$\Rightarrow S \rightarrow aA/bB/b$
 $A \rightarrow aA/bB.$
 $B \rightarrow b.$

$\rightarrow S \rightarrow aA/1B/0C$
 $A \rightarrow aA/0$
 $B \rightarrow xOB.$

$\Rightarrow S \rightarrow 0A$
 $A \rightarrow aA/0.$



$\rightarrow S \rightarrow 0A/1B$
 $A \rightarrow 1B/1$
 $B \rightarrow 0A/0$

$\Rightarrow S \rightarrow XA/YB$
 $A \rightarrow YB/1$
 $B \rightarrow XA/0$
 $X \rightarrow 0$
 $Y \rightarrow 1.$

$\rightarrow S \rightarrow ABaB$
 $A \rightarrow aA/a$
 $B \rightarrow bB/b$

$\Rightarrow S \rightarrow ABxB$
 $A \rightarrow XA/a$
 $B \rightarrow YB/b$
 $X \rightarrow a$
 $Y \rightarrow b.$

$S \rightarrow P0$
 $P \rightarrow AB, 0 \rightarrow XB$
 $A \rightarrow XA/a, B \rightarrow YB/b$
 $X \rightarrow a, Y \rightarrow b.$

$\rightarrow S \rightarrow 1A/0B$
 $A \rightarrow 0S/0$
 $B \rightarrow 1S/1/\epsilon$

Remove $S \rightarrow 1A/0B/0$
 $A \rightarrow 0S/0$
 $B \rightarrow 1S/1$
 ϵ -Prod!

no, unit, useless, dead prod! so go for step-4.

$S \rightarrow XA/YB/0$
 $A \rightarrow YB/0$
 $B \rightarrow XA/1$
 $X \rightarrow 1$
 $Y \rightarrow 0.$

CFG to CNF.

Page No. 84.

29/08/25
Friday

Page No. 85.

→ $A \rightarrow CAC/AB/B/e$
 $B \rightarrow 00/e$
 $C \rightarrow 11/e$

Remove ϵ -
Production.

→ $A \rightarrow CAC/AB/B/CC/A/$
 $CA/AC/C/11.$

Augmented $A' \rightarrow AU\{e\}$
 Prod?

Use when
 ϵ is present in
 initial state.

$B \rightarrow 00$
 $C \rightarrow 11$

$X \rightarrow 00, Y \rightarrow 11$

$A \rightarrow CAC/AB/XX/CC/$
 $CA/AC/YY$

$B \rightarrow XX$
 $C \rightarrow YY$

$X \rightarrow 0, Y \rightarrow 1, W \rightarrow AC$

$A \rightarrow CW/AB/XX/CC/CA/$
 AC/YY

$B \rightarrow XX$
 $C \rightarrow YY$

→ $S \rightarrow aSBcD/BC$
 $A \rightarrow AbCd/a$
 $B \rightarrow CBA/e$
 $C \rightarrow c/e$
 $D \rightarrow d$

Remove ϵ -
 Prod? $\Rightarrow$ $S \rightarrow aSBcD/BC/$
 $aScD/B/C/e$
 $A \rightarrow AbCd/a/Abd$
 $B \rightarrow CBA/BA/A/CA$
 $C \rightarrow c$
 $D \rightarrow d$

Remove Unit
 Production.

$S \rightarrow aSBcD/BC/aScD/CBA/BA/$
 $CA/AbCd/a/Abd/c$

$A \rightarrow AbCd/a/Abd$
 $B \rightarrow CBA/BA/CA/AbCd/a/Abd$
 $C \rightarrow c$
 $D \rightarrow d$

Terminal to
 variable conversion.

$x \rightarrow a, y \rightarrow b$

$a \rightarrow X, b \rightarrow Y, c \rightarrow Z, d \rightarrow W$

$S \rightarrow XSBZD/$

$S \rightarrow XSBcD/BC/XScD/CBA/BA/CA/AYC/$
 $a/AyD/c$

$A \rightarrow AYCD/a/AYD$

$B \rightarrow CBA/BA/CA/AYCD/a/AYD$

$C \rightarrow c$

$D \rightarrow d$

$P \rightarrow XS, Q \rightarrow BC,$
 $R \rightarrow CB, M \rightarrow AY, Z \rightarrow CD, N \rightarrow PQ$

$S \rightarrow PCD/BC/PCD/C/RA/BA/CA/$
 $MD/a/MD$
 $A \rightarrow MCD/a/MD$
 $B \rightarrow RA/BA/CA/MCD/a/MD$



$S \rightarrow ND/BC/PZ/C/RA/BA/CA/MZ/a/MD$
 $A \rightarrow MZ/a/MD$
 $B \rightarrow RA/BA/CA/MZ/a/MD$

$\rightarrow S \rightarrow i(B)SK/p$
 $K \rightarrow OS/e$
 $B \rightarrow t/f$

Gabbage Normal form.
 * CFG to GNF :-

GNF:
 $S \rightarrow TV^*$

CNF:
 $S \rightarrow VV$

- Step-1 :- Convert CFG to CNF.
- Step-2 :- Remove Left Recursion
- Step-3 :- Convert first symbol as a Terminal.

Eg. :- $A \rightarrow ABa/Ba/a$
 $\alpha \quad B_1 \quad B_2$
 Left Recursion: $B \rightarrow a/b$

$\Rightarrow A \rightarrow BA'/aA' \Rightarrow A \rightarrow BAA'/aA'$
 $A' \rightarrow BAA'/e$

$\rightarrow S \rightarrow TBX/abX$
 $B \rightarrow BX/a$
 $X \rightarrow aAb/e$
 $T \rightarrow bT/e$

First convert to CNF
 Remove ϵ -prod.

X

$S \rightarrow TBX/abX/bBX/$
 $B \rightarrow BX/a/ab$
 $X \rightarrow aAB/ab$
 $T \rightarrow bT/b$

$\rightarrow S \rightarrow TBX/abX$
 $B \rightarrow BX/a$
 $X \rightarrow aAb/E$
 $A \rightarrow bT/E$
 $T \rightarrow b$

$S \rightarrow TBX/abX/TB$
 $B \rightarrow BX/a/B$
 $X \rightarrow aAb/ab$
 $A \rightarrow bT/b$
 $T \rightarrow b$

$\Downarrow$ Remove unit prod.

$S \rightarrow TBX/abX/TB/ab$
 $B \rightarrow BX/a$
 $X \rightarrow aAb/ab, A \rightarrow bT/b, T \rightarrow b$

$\Downarrow$ convert variable to

$S \rightarrow TBX/wTX/TB/wT$
 $B \rightarrow BX/a$
 $X \rightarrow wAT/wT$
 $A \rightarrow TT$
 $T \rightarrow b, w \rightarrow a$

$\Downarrow$

$S \rightarrow TP/RX/TB/wT$
 $B \rightarrow BX/a$
 $X \rightarrow wQ/wT$
 $A \rightarrow TT$
 $T \rightarrow b, w \rightarrow a, P \rightarrow BX, Q \rightarrow AT, R \rightarrow wT$

$S \rightarrow TP/RX/TB/wT$
 $B \rightarrow aB'$
 $B' \rightarrow XB'/E$
 $X \rightarrow wQ/wT$
 $A \rightarrow TT, T \rightarrow b, w \rightarrow a, P \rightarrow BX, Q \rightarrow AT, R \rightarrow wT$

$\Downarrow$

$S \rightarrow TP/RX/TB/wT$
 $B \rightarrow aB'/a$
 $B' \rightarrow XB'/wQ/wT$
 $X \rightarrow wQ/wT$
 $A \rightarrow TT, T \rightarrow b, w \rightarrow a, P \rightarrow BX, Q \rightarrow AT, R \rightarrow wT$

$\Downarrow$ convert first symbol as Terminal.

$S \rightarrow bP/aTX/bB/aT$
 $B \rightarrow aB'/a$
 $B' \rightarrow aQB'/aQ/aT/aTB'$
 $X \rightarrow aQ/aT$
 $A \rightarrow bT$
 $T \rightarrow b$
 $w \rightarrow a$
 $P \rightarrow aB'x/ax$
 $Q \rightarrow bTT$
 $R \rightarrow aT$